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Original Contribution

BMI Trajectories in Late Middle Age, Genetic Risk, and Incident Diabetes in Older Adults: Evidence From a 26-Year Longitudinal Study

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This study investigated the association between body mass index (BMI) trajectories in late middle age and incident diabetes in later years. A total of 11,441 participants aged 50–60 years from the Health and Retirement Study with at least 2 self-reported BMI records were included. Individual BMI trajectories representing average BMI changes per year were generated using multilevel modeling. Adjusted risk ratios (ARRs) and 95% confidence intervals (95% CIs) were calculated. Associations between BMI trajectories and diabetes risk in participants with different genetic risks were estimated for 5,720 participants of European ancestry. BMI trajectories were significantly associated with diabetes risk in older age (slowly increasing vs. stable: ARR = 1.31, 95% CI: 1.12, 1.54; rapidly increasing vs. stable: ARR = 1.5, 95% CI: 1.25, 1.79). This association was strongest for normal-initial-BMI participants (slowly increasing: ARR = 1.34, 95% CI: 0.96, 1.88; rapidly increasing: ARR = 2.06, 95% CI: 1.37, 3.11). Participants with a higher genetic liability to diabetes and a rapidly increasing BMI trajectory had the highest risk for diabetes (ARR = 2.15, 95% CI: 1.67, 2.76). These findings confirmed that BMI is the leading risk factor for diabetes and that although the normal BMI group has the lowest incidence rate for diabetes, people with normal BMI are most sensitive to changes in BMI.

BMI trajectory; longitudinal study; new-onset diabetes

Abbreviations: ARR, adjusted risk ratio; BMI, body mass index; CI, confidence interval; HR, hazard ratio; HRS, Health and Retirement Study; PRS, polygenic risk score.

Diabetes has resulted in a major global disease burden in aging societies. The prevalence of diabetes is projected to increase to 11.8%, with 29.6 million patients in the United States by 2030 (1, 2). People aged over 60 years contribute most to the population with diabetes, with an increase in the prevalence of diabetes among older age groups compared with younger age groups (2–4).

Elevated body mass index (BMI) is the leading risk factor for type 2 diabetes in most cases (5). While most studies have acquired the BMI data of participants based on single or limited measurements, some studies have investigated the association of long-term BMI changes with type 2 diabetes. Bao et al. (6) revealed that weight gain was significantly associated with type 2 diabetes in women with a history of gestational diabetes. However, controversy exists regarding whether BMI changes affect diabetes incidence depending on the attained BMI. Jacobs-van der Bruggen et al. (7) suggested that weight change has no effect on diabetes risk beyond the effects on attained BMI, while Black et al. (8) had the opposite findings. Nano et al. (9) classified BMI trajectories into 3 categories, including progressive overweight, progressive weight loss, and persistently high BMI, before the diagnosis of type 2 diabetes. They pointed out that the majority of people may miss the opportunity for screening efforts, with only high-risk individuals obtaining attention. Thus, prevention strategies should also be addressed to those with increasing BMI trajectories instead of focusing only on those with persistently high BMI.

To our knowledge, little evidence has been provided on the association between BMI trajectories in middle age and diabetes risk in older age. By using data from the Health and Retirement Study (HRS) cohort, a large representative longitudinal cohort in the United States, we were able to use repeated self-reported BMI from individuals between the ages of 50 and 60 years to generate BMI trajectories in late middle age for each participant by constructing a multilevel model. We aimed to investigate the association between BMI trajectories in late middle age and the risk of diabetes in older age and to explore whether this association was independent of initial BMI in late middle age and the inherent polygenic risk score (PRS) of the individual.

METHODS

Study design and cohort selection

The analysis was based on HRS, which is sponsored by the National Institute on Aging and conducted by the University of Michigan. The HRS is a longitudinal survey that collected data for approximately 42,233 participants aged 50 years or older in the United States once every 2 years from 1992 to 2020. For each wave, both extensive health information and sociodemographic characteristics were collected for each individual. Since 2006, genetic data has also been included in the data collection process (10). The process of the current study is presented in Web Figure 1A (available at https://doi.org/10.1093/aje/kwad080).

In the present study, we included all participants taking part in the survey between 1992 and 2018. Participants were excluded if they: 1) had an initial BMI less than 18.5 (n = 281); 2) were recruited after 60 years of age (n = 16,013); 3) had extreme BMI records (bottom 1% and top 1%, n = 235); 4) had fewer than 2 BMI records (n = 4,599); 5) had physician-diagnosed diabetes before 60 years of age (n = 4,257); 6) did not have any diabetes information after they were 60 years old (n = 5,298); or 7) had any missing information for covariates (n = 109).

Measures of exposures and outcomes

A self-reported weight in kilograms was recorded for each wave, and a self-reported height was obtained only from new respondents, considering that the height of elderly participants rarely changed. BMI was calculated as weight (kilograms) divided by the square of height (meters). The age of each participant at each interview was derived from the date of the interview and the date of birth. The period of late middle age was defined as an age of 50–60 years, while older age was defined as an age greater than 60 years. The BMI trajectories of individuals between the ages of 50 and 60 years were extracted as a continuous variable from a multilevel model using a random intercept and slope (Web Figure 1), and individual-level and initial BMI levels (normal, overweight, and obese) were included in the model. The initial BMI was defined as the earliest BMI recorded between 50 and 60 years of age and was recategorized as “normal” (18.51–25.00), “overweight” (25.01–30.00), or “obese” (>30.00). Information on diabetes status was acquired at each wave using the question “Has a doctor ever told you that you have diabetes or high blood sugar?” According to previous studies, BMI measurements based on self-reported weight and height could be used to estimate the health risks with variations in BMI once the sociodemographic characteristics were adjusted for (11). Additionally, self-reported diabetes could identify participants with diabetes with high sensitivity and specificity (12).

Covariates

Covariates included sex (female, male), race (White, Black/African American, and other), marital status (married, never married, separated/divorced, and widowed), years of education, year of study enrollment (1990–2000, 2001–2010, and 2011–2020), age when the first BMI record occurred, smoking status (current, ever, never), ever drinking alcohol (yes and no), and long-standing illness status (i.e., hypertension, heart disease, stroke, arthritis, lung disease, and psychiatric disorders). The age when the first BMI record occurred was defined as the start of the BMI records, as the period of trajectory differed from individual to individual. Other covariates were derived from the latest record before the participant was 60 years of age.

Polygenic risk score

Genotyping was conducted by the Center for Inherited Disease Research (CIDR) in 2011, 2012, and 2015 (RC2 AG0336495 and RC4 AG039029). Genotype data for over 19,000 HRS participants were obtained using the Illumina HumanOmni2.5 BeadChip array (HumanOmni2.5-4v1, HumanOmni2.5-8v1, HumanOmni2.5-8v1.1; Illumina, San Diego, California), which measures ∼2.4 million single nucleotide polymorphisms (SNPs). Individuals were excluded if they had missing call rates >2%, SNPs with call rates <98%, Hardy-Weinberg equilibrium P values <0.0001, chromosomal anomalies, and first-degree relatives in the HRS.

As type 2 diabetes accounts for 91.2% (90.4%, 92.1%) of diabetes diagnoses among US adults (13), the PRS for type 2 diabetes, which represents a genetic predisposition to diabetes, could reflect the genetic risk for diabetes in US adults to a great extent. We performed a secondary analysis that included only participants whose self-reported racial/ethnic background was White and those with genetic information collected. The PRS of diabetes was calculated for 5,720 participants using the results of a genome-wide association study meta-analysis from the Diabetes Genetics Replication and Meta-analysis (DIAGRAM) Consortium (14). Participants were reclassified as low risk, intermediate risk, and high risk by tertiles according to the distribution of PRSs in the general cohort. The first 5 principal components were also included in the secondary analysis to adjust for possible population stratification.

Statistical analysis

The analysis of the association between BMI trajectory and incidence of diabetes followed 3 steps: the evaluation of BMI trajectory, description and classification for BMI trajectory, and estimation of the association between BMI trajectory and incident diabetes.

Step 1. We used individual-level data of participants without diabetes before the age of 60 years to construct and fit a multilevel linear regression model and estimate the relationship between BMI and time (age at the time of the record). BMI itself was included as the dependent variable, while the intercept and time were treated as random effects at the individual level. We also adjusted for confounders, including initial BMI group, sex, and age when the first record occurred, as fixed effects in the model. The estimated model parameter for age at the time of the record was extracted as the tendency of BMI change for each participant, the corresponding code used to derive BMI trajectory is presented in Web Appendix 1. Here, we defined a positive tendency of BMI change as an increasing BMI trajectory and a negative tendency of BMI change as a decreasing BMI trajectory.

Step 2. A generalized linear model (with a log link and Poisson distribution) was fitted to estimate the association of the BMI trajectory with the incidence of diabetes. A restricted cubic spline with 4 knots was constructed to explore whether any nonlinear association existed (Figure 1). Depending on the distribution of the BMI trajectory, the BMI trajectory was classified as "rapidly decreasing" (bottom 20% of the total decreasing BMI trajectory), "slowly decreasing," "stable" (top 10% of the total decreasing BMI trajectory and bottom 10% of the increasing BMI trajectory), "slowly increasing," and "rapidly increasing" (top 20% of the total increasing BMI trajectory) (Web Figure 2). The fluctuation of BMI for different BMI trajectories was calculated by the range of the BMI records to reveal the degree of BMI change in different groups.

Step 3. We fitted a general linear model adjusting for sex, race, years of education, marital status, age when the first record occurred, year of study enrollment, smoking status, ever drinking alcohol, long-standing illness status, and initial BMI group. We also performed a trend test to examine the dose-response relationship between BMI trajectory and diabetes risk using the continuous BMI trajectory. Stratified analyses were also performed depending on the initial BMI groups of the participants. Due to the relatively small sample size in the secondary analysis and subgroups, we merged the BMI trajectories into "stable," "falling" ("rapidly decreasing," "slowly decreasing"), and "rising" ("rapidly increasing," "slowly increasing"). Furthermore, we tested the interaction effect between the initial BMI groups and the BMI trajectory by comparing the fitted model with and without the interaction term. Subsequent analysis was performed if heterogeneity was found between the initial BMI group and BMI trajectory.

Finally, we examined whether the BMI trajectories were associated with diabetes risk in participants with a genetic predisposition to type 2 diabetes. We first adjusted for the PRS as continuous variable and then tested the association between the BMI trajectories and incident diabetes among participants stratified by PRS.

Some sensitivity analyses were also performed. First, we restricted the sample to participants with at least 3 recorded BMIs in late middle age, as 9,047 participants had at least 3 BMI records. Second, we reassessed the association of the BMI trajectories and risk for diabetes by stratifying the participants by the mean BMI group instead of the initial BMI group before the age of 60 years to estimate whether the association between BMI trajectory and incident diabetes was independent of overall BMI status. Third, because the HRS recruited new participants to refresh the cohort, the participants might have different follow-up intervals, and we repeated the analyses by using a Cox proportional hazards model to evaluate hazard ratios (HRs) with 95% confidence intervals (CIs). The timescale was defined as the difference between the age of 60 years and the age at the interview in which diabetes was first reported or the latest interview, whichever came first. Finally, we relaxed the time restriction for BMI measurements, and the BMI trajectory was constructed in the other way that it includes all BMI records prior diabetes diagnosis to define a more general BMI trajectory. The multilevel model was fitted by using the R (R Foundation for Statistical Computing, Vienna, Austria) package lme4. *P* values were 2-sided, with statistical significance set as less than 0.05. All analyses were performed using R version 3.6.3.

RESULTS

Population characteristics

Between 1992 and 2018, a total of 11,441 participants who did not have diabetes or high blood sugar at the age of 60 were included in the main analysis. The comparison among participants who were excluded because of a lack of follow-up information (e.g., participants younger than 60 years at their last interview) and those who were included in our cohort are shown in Web Table 1. Participants who were excluded due to unavailable data for a diabetes diagnosis were more likely to be recruited later, so relatively less information was collected during follow-up. Among the included participants, a total of 2,308 incident diabetes patients were identified after 60 years of age. As presented in Table 1, compared with the participants who had a stable BMI trajectory, participants with a falling BMI trajectory were more likely to be overweight or obese initially, participants with a rising BMI trajectory were more likely to have a normal BMI, and participants with a rapidly increasing BMI were more likely to be obese and have other long-standing illnesses at 60 years of age.

As the classification of BMI trajectory was derived by its distribution in the total participants, it was found that the participants grouped in the rapidly increasing or decreasing BMI trajectory groups experienced greater volatility of weight during late middle age (Web Table 2). Compared with participants who had a falling BMI trajectory, those who were classified as having a rising trajectory had a wider fluctuation.

Table 1. The Baseline Characteristics[a] of Participants at 60 Years of Age in Relation to Body Mass Index Trajectory During Late Middle Age in a Subset (n = 11,441) of the Health and Retirement Study, United States, 1992–2018

Characteristic	Stable (*n* = 1,145)		Falling (*n* = 4,828)				Rising (*n* = 5,468)			
			Rapidly Decreasing (*n* = 1,073)		Slowly Decreasing (*n* = 3,755)		Slowly Increasing (*n* = 4,253)		Rapidly Increasing (*n* = 1,215)	
	No.	%	No.	%	No.	%	No.	%	No.	%
Initial BMI[b] group										
Normal	255	38.3	1,195	27.8	289	35.9	2,005	40.0	124	19.0
Overweight	316	47.5	1,655	38.5	378	46.9	2,203	43.9	143	21.9
Obesity	94	14.1	1,451	33.7	139	17.2	809	16.1	385	59.0
Sex										
Female	358	53.8	2,451	57.0	424	52.6	2,669	53.2	478	73.3
Male	307	46.2	1,850	43.0	382	47.4	2,348	46.8	174	26.7
Race										
Black/African American	114	17.1	710	16.5	122	15.1	834	16.6	155	23.8
White	514	77.3	3,327	77.4	633	78.5	3,879	77.3	454	69.6
Other	37	5.6	264	6.1	51	6.3	304	6.1	43	6.6
Year of enrollment										
1990–2000	314	47.2	1,941	45.1	393	48.8	2,249	44.8	191	29.3
2001–2010	206	31.0	1,281	29.8	232	28.8	1,572	31.3	243	37.3
2011–2019	145	21.8	1,079	25.1	181	22.5	1,196	23.8	218	33.4
Marital status[c]										
Never married	28	4.2	192	4.5	36	4.5	226	4.5	39	6.0
Partnered	493	74.1	3,140	73.0	590	73.2	3,793	75.6	438	67.2
Separated/Divorced	99	14.9	693	16.1	141	17.5	694	13.8	113	17.3
Widowed	45	6.8	276	6.4	39	4.8	304	6.1	62	9.5
Years of education[d]	13.0 (3.02)		12.9 (3.04)		13.1 (2.98)		12.9 (3.02)		12.6 (2.95)	
Age when the first record occurred[d]	54.1 (2.21)		54.0 (2.18)		54.0 (2.20)		53.7 (2.17)		53.0 (2.13)	
Ever drinking alcohol[c]										
No	275	41.4	1,720	40.0	316	39.2	1,995	39.8	361	55.4
Yes	390	58.6	2,581	60.0	490	60.8	3,022	60.2	291	44.6

Table continues

Table 1. Continued

Characteristic	Stable (n = 1,145)		Falling (n = 4,828)				Rising (n = 5,468)			
			Rapidly Decreasing (n = 1,073)		Slowly Decreasing (n = 3,755)		Slowly Increasing (n = 4,253)		Rapidly Increasing (n = 1,215)	
	No.	%	No.	%	No.	%	No.	%	No.	%
Smoking status[c]										
Current	151	22.7	1,068	24.8	163	20.2	945	18.8	76	11.7
Ever	230	34.6	1,524	35.4	323	40.1	1,990	39.7	274	42.0
Never	284	42.7	1,709	39.7	320	39.7	2,082	41.5	302	46.3
Long-standing illness at baseline[c]										
No	244	36.7	1,458	33.9	285	35.4	1,526	30.4	87	13.3
Yes	421	63.3	2,843	66.1	521	64.6	3,491	69.6	565	86.7

Abbreviations: BMI, body mass index; SD, standard deviation;
[a] All data shown are percentages unless otherwise specified.
[b] Calculated as weight (kg)/height (m)^2. Normal: 18.51–25.00; overweight: 25.01–30.00; or obesity: >30.00.
[c] Based on the most recent assessment before or at 60 years of age.
[d] Values are expressed as mean (standard deviation).

Association between BMI trajectory and diabetes

The incidence of type 2 diabetes is presented in Web Figure 3. Restricted cubic splines adjusted for the same covariates in the multivariable model were used to visualize the relationship of the BMI trajectories with diabetes incidence in older age to demonstrate whether a nonlinear association existed. Generally, to a certain extent, the rising BMI trajectory was associated with an increased risk of diabetes; however, such a linear association was no longer significant at a rapidly decreasing value (P for nonlinearity = 0.005, Figure 1A). Stratified by BMI subgroups, the association between continuous BMI trajectory and diabetes incidence in older age in all subgroups is presented in Figure 1B–D.

In multivariable analyses, our results demonstrated that the risk of diabetes was elevated in participants with a rising BMI trajectory (slowly increasing vs. stable: adjusted risk ratio (ARR) = 1.31, 95% CI: 1.12, 1.54; rapidly increasing vs. stable: ARR = 1.5, 95% CI: 1.25, 1.79; P for trend < 0.001) (Table 2). The test for interaction demonstrated a strong heterogeneity of BMI subgroups on the association between BMI trajectories and diabetes risk (P for interaction < 0.001). As shown in Table 2, significant associations were found in the groups with normal (slowly increasing vs. stable: ARR = 1.34, 95% CI: 0.96, 1.88; rapidly increasing vs. stable: ARR = 2.06, 95% CI: 1.37, 3.11) and overweight (slowly increasing vs. stable: ARR = 1.42, 95% CI: 1.14, 1.77; rapidly increasing vs. stable: ARR = 1.83, 95% CI: 1.39, 2.41) BMI. In the obesity group, although a significant association was found when the quantified BMI trajectory was included in the model (P for trend = 0.003), the association was attenuated when the stable group was defined as the reference group.

Association between BMI trajectory and diabetes after adjustment for genetic risk

A total of 5,720 participants were included in the secondary analyses, in which 1,040 participants developed incident diabetes. The basic characteristics are presented in Web Table 3. The comparison of characteristics between participants who were excluded and those who were included is listed in Web Table 4. As there was no evidence of nonlinearity between the PRS and diabetes after adjusting for the initial BMI group (P for nonlinearity = 0.519), the PRS for type 2 diabetes was included as a continuous variable. The association between the BMI trajectory and type 2 diabetes remained in all participants (slowly increasing vs. stable: ARR = 1.41, 95% CI: 1.11, 1.79; rapidly increasing vs. stable: ARR = 1.67, 95% CI: 1.27, 2.21), in the groups with normal (rapidly increasing vs. stable: ARR = 2.64, 95% CI 1.42, 4.92) and overweight (slowly increasing vs. stable: ARR = 1.56, 95% CI: 1.12, 2.17; rapidly increasing vs. stable: ARR = 1.95, 95% CI: 1.28, 2.98) BMI after additionally adjusting for the PRS and the first 5 principal components (Web Table 5).

The categorized PRS was derived by tertiles. The tendency of risk of diabetes among participants with different BMI trajectories and genetic liability for diabetes is presented in Figure 2A. After stratifying participants by

Table 2. The Association Between Body Mass Index Trajectories in Late Middle Age and Risk of Diabetes in Older Age in a Subset of the Health and Retirement Study, United States, 1992–2018

Study Population and BMI[a] Trajectory	No. of Participants	No. of Cases	ARR[b]	95% CI	*P* Value	*P* for Trend[c]
Overall						
Stable	1,145	192	1.00	Referent		<0.001
Rapidly decreasing	1,073	221	0.88	0.72, 1.08	0.223	
Slowly decreasing	3,755	652	0.97	0.83, 1.14	0.713	
Slowly increasing	4,253	898	1.31	1.12, 1.54	0.001	
Rapidly increasing	1,215	345	1.50	1.25, 1.79	<0.001	
Normal						
Stable	434	40	1.00	Referent		<0.001
Rapidly decreasing	136	7	0.61	0.27, 1.37	0.23	
Slowly decreasing	1,241	84	0.73	0.50, 1.07	0.11	
Slowly increasing	1755	224	1.34	0.96, 1.88	0.09	
Rapidly increasing	302	60	2.06	1.37, 3.11	0.001	
Overweight						
Stable	537	97	1.00	Referent		<0.001
Rapidly decreasing	248	31	0.77	0.51, 1.16	0.207	
Slowly decreasing	1,654	286	0.98	0.78, 1.23	0.847	
Slowly increasing	1879	475	1.42	1.14, 1.77	0.002	
Rapidly increasing	377	116	1.83	1.39, 2.41	<0.001	
Obese						
Stable	174	55	1.00	Referent		0.003
Rapidly decreasing	689	183	0.81	0.60, 1.10	0.175	
Slowly decreasing	860	282	0.99	0.74, 1.32	0.933	
Slowly increasing	619	199	1.06	0.78, 1.43	0.714	
Rapidly increasing	536	169	1.14	0.84, 1.54	0.414	

Abbreviation: ARR: adjusted risk ratio; BMI, body mass index; CI: confidence interval.
[a] Calculated as weight (kg)/height (m)^2. Normal: 18.51–25.00; overweight: 25.01–30.00; or obesity: >30.00.
[b] The multivariable model was adjusted for initial BMI group, sex, race, age when the first BMI record occurred, years of education, marital status, year of enrollment, smoking status, ever drinking alcohol, and long-standing illness status.
[c] The test for trend was conducted by using the continuous BMI trajectory as an independent factor.

PRS, significant associations between BMI trajectory and incident diabetes were found in participants with lower or moderate genetic liability for diabetes (Figure 2B). We also tested the association between incident diabetes and risk groups defined by genetic liability and BMI trajectory in the different initial BMI subgroups (Web Figures 4–6).

Subgroup and sensitivity analyses

Subgroup analyses were conducted after stratifying by risk factors, and the potential heterogeneity was estimated. Heterogeneity was only found between BMI and the BMI trajectory for the risk of diabetes (Web Table 6). With a merged BMI trajectory in subgroup analysis, a significantly protective association of the falling BMI trajectory was found in the initial-normal group (falling vs. stable: ARR = 0.7, 95% CI: 0.52, 0.96; rising vs. stable: ARR = 1.5, 95% CI: 1.17, 1.94). Only a marginally significant association was found in the initial-overweight group (falling vs. stable: ARR = 0.85, 95% CI: 0.72, 1.02; rising vs. stable: ARR = 1.35, 95% CI: 1.15, 1.59). In the sensitivity analysis, the association between the BMI trajectory and diabetes remained (Web Tables 7–10). In the analysis, which was restricted to participants with at least 3 BMI records, participants with a rapidly decreasing BMI trajectory were less likely to develop diabetes (ARR = 0.71, 95% CI: 0.51, 1, Web Table 7). However, the association was partly attenuated in the obesity group when we reclassified the BMI group by mean BMI records (rapidly increasing vs. stable: ARR = 1.17, 95% CI: 0.98, 1.41; Web Table 8). The association between BMI and incident diabetes in older individuals in the main analysis and sensitivity analysis is presented in Web Table 11.

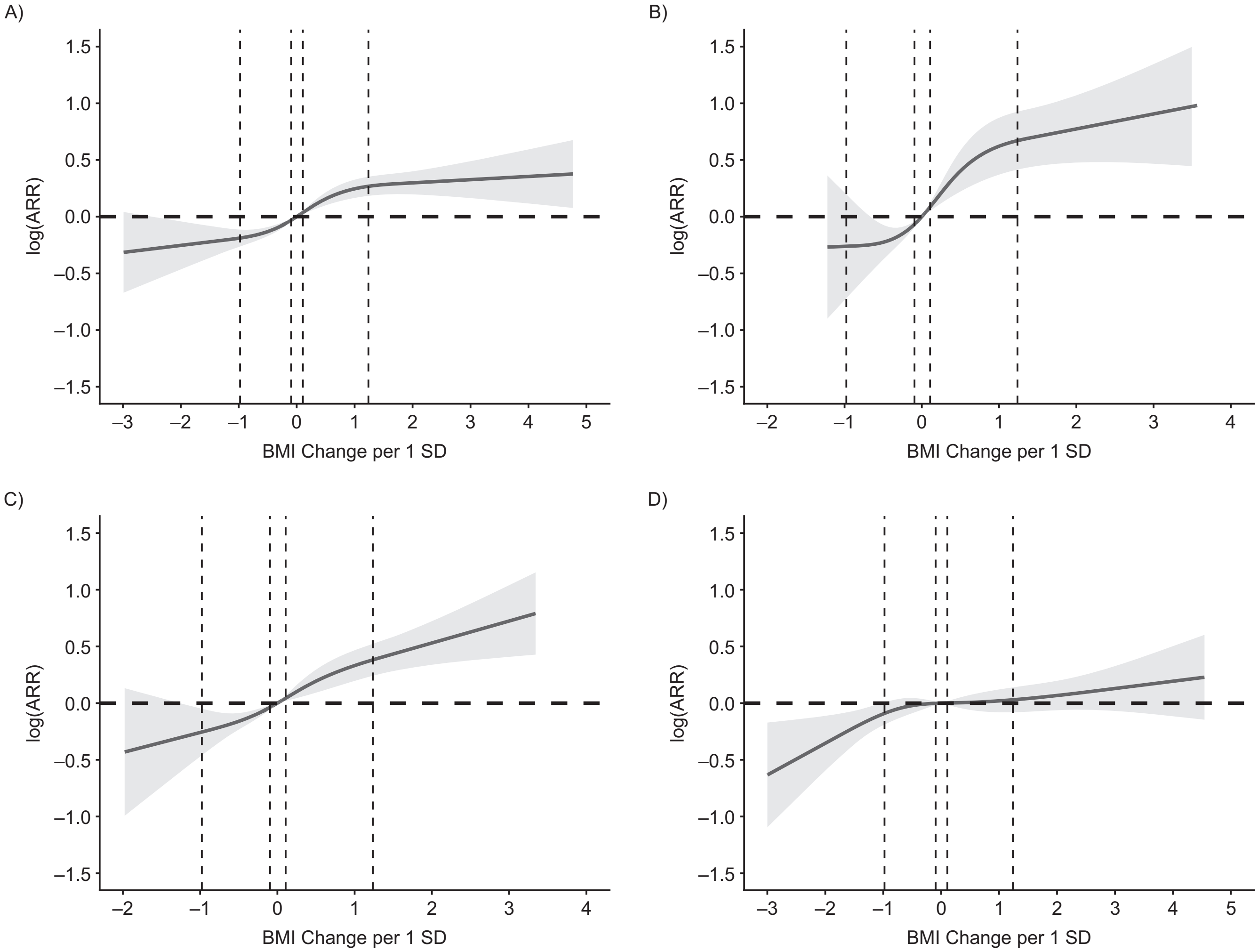

Figure 1. Restricted cubic splines demonstrated the nonlinear association between body mass index (BMI) trajectories in late middle age and diabetes in older age, Health and Retirement Study, United States, 1992–2018. A) The overall sample (*P* for nonlinearity = 0.004); B) subgroup with normal (*P* for nonlinearity = 0.019); C) overweight (*P* for nonlinearity = 0.441); and D) obesity (*P* for nonlinearity = 0.182). Models additionally adjusted for sex, race, coefficient of variance of BMI records, years of education, marital status, year of enrollment, smoking status, ever drinking alcohol, and long-standing illness status. Depending on the distribution of the BMI trajectory, the BMI trajectory was classified as "rapidly decreasing" (bottom 20% of the total decreasing BMI trajectory, the left of the first vertical line), "slowly decreasing" (between the first and second vertical line), "stable" (top 10% of the total decreasing BMI trajectory and bottom 10% of the increasing BMI trajectory, between the second and third vertical line), "slowly increasing" (between the third and fourth vertical line), and "rapidly increasing" (top 20% of the total increasing BMI trajectory, the right of the fourth vertical line).

DISCUSSION

In general, BMI during middle age was strongly associated with diabetes incidence in older age, even after adjusting for genetic liability to diabetes. The BMI trajectory in late middle age was significantly associated with the risk of diabetes in older age, and the association persisted even in the normal BMI group. Furthermore, we found that the PRS of diabetes has an additive effect rather than an interactive effect on the association between BMI trajectories and later diabetes risk.

BMI changes have been linked to a series of diseases in later life. Zheng et al. (15) reported that weight gain in early to middle adulthood could significantly increase the risk of multiple major chronic diseases, including type 2 diabetes, cardiovascular disease, and obesity-related cancer. Another study based on the Framingham Heart Study cohort suggested that life-long BMI trajectories were associated with mortality risk (16). For diabetes, Bao et al. (6) demonstrated that each 5-kg increase in weight was associated with a 27% increased risk of type 2 diabetes (HR = 1.27, 95% CI: 1.04, 1.54) among women with a history of gestational diabetes. Evidence showed that young and middle-aged adults whose weight changed from obese to nonobese had a reduced risk of diabetes compared with those who remained obese (HR = 0.33, 95% CI: 0.14, 0.76) (17). However, with a small sample

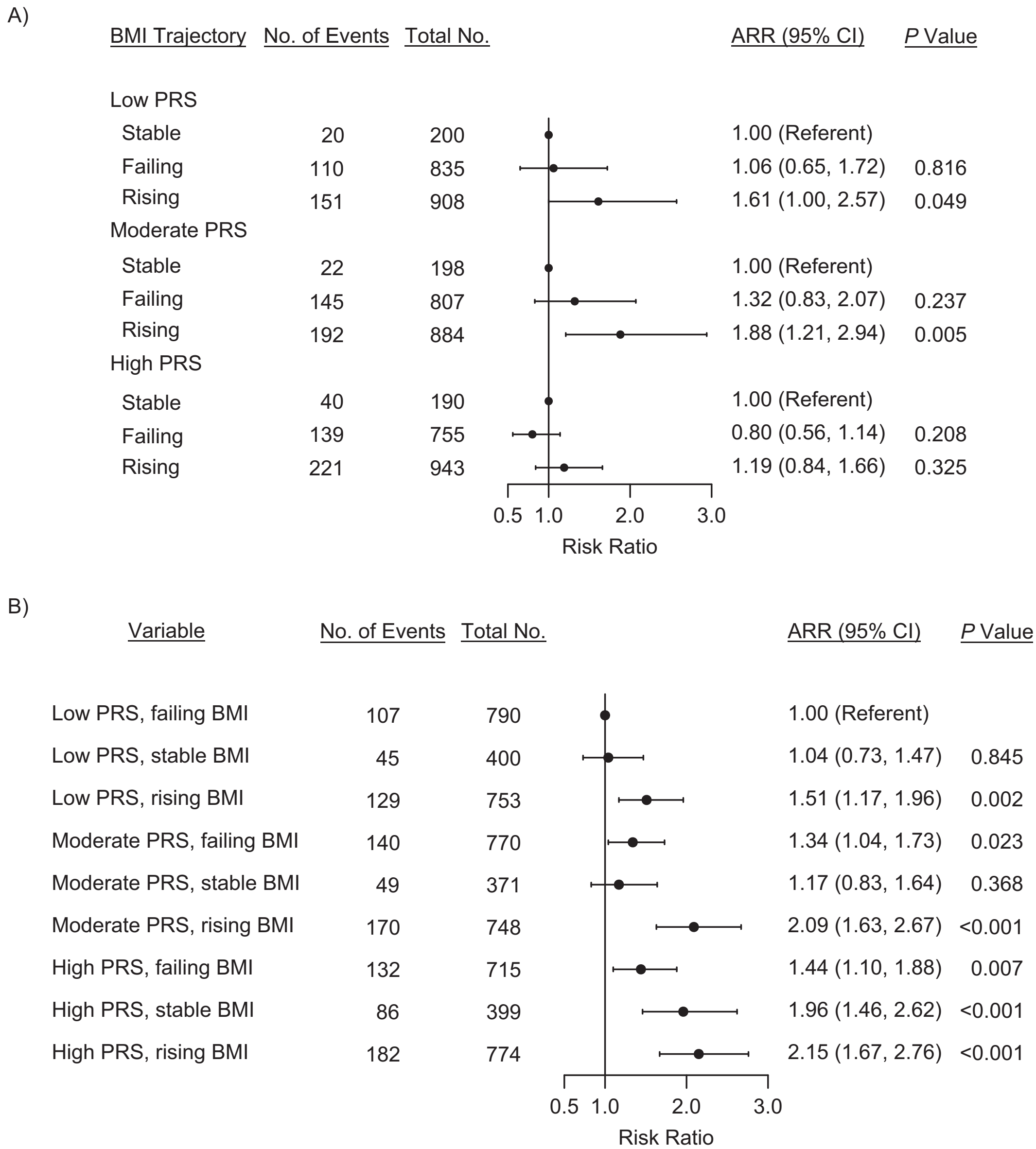

Figure 2. Risk of diabetes in older age according to body mass index (BMI) trajectories in middle age and polygenic risk score (PRS), Health and Retirement Study, United States, 1992–2018. A) Stratified by type 2 diabetes; B) reclassified groups by PRS and BMI trajectory. Models adjusted for sex, race, coefficient of variance of BMI records, years of education, marital status, year of enrollment, smoking status, ever drinking alcohol, long-standing illness status, and the first 5 principal components.

size in the "obese to nonobese" group, part of their results should be interpreted with caution in light of potentially insufficient statistical power.

Our results were not only in line with but also further extended the results of previous studies, suggesting that, compared with participants with a stable BMI trajectory, participants with a slowly increasing BMI trajectory might have a 31% higher risk for developing diabetes, and participants with a rapidly increasing BMI might experience a 50% higher risk for diabetes, regardless of initial BMI. We found that the association between BMI trajectories and diabetes risk in middle age and later in life persisted in individuals with normal BMI but not in obese individuals, showing that increasing BMI may play a role in the development of later-life diabetes in individuals with a normal initial BMI, despite the important role that initial BMI played in late middle age. A possible explanation is that most of the obese participants already had diabetes before 60 years of age (Web Table 12); in contrast, participants with normal BMI and a rising trajectory might be more likely to be in the overweight or obese groups in their 60s, which leads to an enhanced association with diabetes after 60 years of age.

Our study adds to the body of evidence linking BMI status and incident diabetes (18–20). In the initial BMI–based subgroup, a significant association between weight loss and incident diabetes was only found in participants with normal BMI at baseline. With a similar fluctuation, participants who were in a normal BMI range were more likely to benefit from controlling their BMI than those in an overweight/obese weight range at baseline.

Whether the association between BMI changes and type 2 diabetes depends on BMI is also ambiguous. Jacobs-van der Bruggen et al. (7) estimated the association between 5-year weight change and diabetes risk in the following 5 years and found that weight changes in 5 years had no effect on diabetes incidence when considering attained BMI. In contrast, Black et al. (8) found that for any given BMI, the risk of impaired glucose tolerance increased in those who had gained weight since they were 20 years of age (odds ratio = 1.10 per unit BMI, 95% CI: 1.03, 1.17), and the association was independent of attained BMI. Our study further expands the association between BMI trajectory and incident diabetes by using repeated BMI records.

We also considered the potential effects of the genetic risk of type 2 diabetes on the association between BMI trajectories and type 2 diabetes. The PRS of type 2 diabetes has already been reported to be associated with increased mortality risk (21) and might be related to hypertension, coronary artery disease, and stroke (22). Schnurr et al. (23) found no significant interaction between PRS and either BMI or unfavorable lifestyles concerning the risk of type 2 diabetes. Similarly, our findings suggested an additive effect rather than an interactive effect of the PRS on the association between BMI trajectories and diabetes.

The main advantage of the present study was that we quantified BMI trajectories by using a multilevel model. Since multiple observations were nested within individuals, the independence assumption needed for traditional statistical analysis was violated. The multilevel model allowed us to maximize data utilization with the ability to handle missing data well, thus making it possible to characterize BMI as a time-varying exposure. In addition, the multilevel model could fit BMI trajectories for each participant and consider the general population at the same time (24). Second, the cohort we used was a nationally representative cohort containing 11,441 participants with repeated self-reported BMI records, which provided sufficient statistical power to interpret our results. Third, we derived covariates from the latest, instead of the first, BMI record to obtain better adjustment for covariates. In addition, by conducting a range of sensitivity analyses and considering the inherent genetic risk of diabetes, the robustness of our results was demonstrated.

The present study should also be interpreted with the understanding of several limitations. First, although self-reported BMI and physically measured BMI are comparable in general in both the United States (25) and other countries (26–28), there is still the possibility of misclassification. Additionally, other self-report varieties may also introduce systematic biases. Second, because we only included participants with available diabetes diagnoses during the follow-up period, some characteristics differed between the included and excluded participants; thus, the results are not representative of the general US population. Finally, although studies have shown that statin users might experience a 10%–45% higher risk of new-onset diabetes mellitus than nonusers (29), data were not collected during our study period. Potent cholesterol-lowering drugs are widely accepted for reducing blood cholesterol concentrations to prevent or abate coronary heart disease, such as 3-hydroxy-3-methylglutaryl coenzyme A reductase inhibitors (statins) (30, 31). We found that participants who had obesity at baseline were more likely to have other long-standing illnesses. Although chronic disease itself was adjusted for in the analysis, the missing information about exact prescription drugs made it impossible to separate the influence of medicine on diabetes. Future studies with more detailed medical and diet records are warranted to assess whether heterogeneity between influencing factors exists.

In conclusion, we found that BMI trajectories in individuals in late middle age were associated with diabetes in older age regardless of genetic predisposition or initial BMI. The present study underlines the necessity of BMI control for the prevention of diabetes in older age and suggests that effective BMI control should be emphasized even in a population with a normal BMI.

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